

2026-yil Holatida Telegram + AI Agent Arxitekturasini: 100 dan 100,000 Foydalanuvchiga Masshtablanadigan Hybrid Arxitektura Qo'llanmasi

Asosiy Arxitektura Tanlovi va Vendor Lock-in Strategiyalari

2026-yil sentyabr holatida Telegram botlari, mini-app'lar va AI agentlar kombinatsiyasini qurish uchun asosiy arxitektura tanlovi ikkita ustuvor ekotizim orasida sodir bo'ladi: "Supabase + Render + Cloudflare" hamjumlashirilgan eko-tizimi yoki "Azure" monolitik platformasi. Har birining o'z afzalliklari, kamchiliklari va uzoq muddatli oqibatlari mavjud bo'lib, bu qaror loyihaning biznes strategiyasiga, jamoaning texnik malakasiga va kelajakdagi rivojlanish kutubxonalariga bog'liq. Ushbu bo'limda ushbu ikkala yo'nalishning chuqur taqqoslamasi, vendor lock-in risklari va har biriga oid arxitektura qarorlari (ADR) tahlil qilinadi.

Arxitektura Tanlovi: Supabase+Render vs. Azure

Bu ikkita arxitektura yondashuvi asosan tez start, minimal konfiguratsiya va minimal operatsion xarajatlar (DevEx) imkoniyati bilan belgilanadi. Supabase'ning PostgreSQL asosidagi backend-xizmatlari (100), Render'dagi Git-orqali-deploy qilish modeli (123) va Cloudflare'ning global CDN va edge computing platformasi (227) bir-biri bilan juda yaxshi integratsiyalashadi. Bu ekotizimning asosiy afzalligi — tezligidir. Jamoa barcha komponentlarni ularning rasmiy interfeyslari (APIs) orqali alohida sozlashi mumkin, bu esa arxitektura qisqa muddatda murakkablikka uchramasdan ishlashi mumkin. Masalan, Telegram bot'ni Render'ga deploy qilish va uning webhook'larini Cloudflare Pages orqali frontend'ga ulash ancha oson va standart metodologiyalar asosida amalga oshiriladi (48, 68).

Biroq, bu erkinlik narxi bor. Eng katta nuqsonlardan biri — vendor lock-in. Agar loyiha bu ekotizimni tanlasa, uni keyinchalik boshqa platformaga ko'chirish qiyin va qimmatga tushishi mumkin. Masalan, agar ma'lumotlar bazasi supabase.org platformasida

joylashsa, uning migratsiyasi PostgreSQL'ning o'ziga tegishli logikaga ehtiyoj bor bo'lgan maxsus sozlamalarni talab qiladi (34). Shuningdek, Cloudflare'ning unikal funksiyalari (masalan, R2'dagi "zero egress fee" modeli 160) yoki Render'dagi Workflows kabi xizmatlar (220) boshqa xosting provayderlarida mavjud bo'lmasligi mumkin. Bundan tashqari, bu ekotizimning narxlari cheklovlar tashkil etilgach keskin oshishi ham muhim hisoblanadi. Masalan, Supabase'ning Pro plani (25/ora)([[120]])vaRender'ningwebservice'lariningProUltravarianti(450/gacha) (117) kabi narxlardan keyin loyihaning oylik xarajatlari sezilarli darajada oshishi mumkin. Shunday qilib, bu arxitektura sinov uchun yoki kichik hajmdagi loyihalar uchun ideal bo'lsa-da, 10,000 dan ortiq foydalanuvchiga yetkazish va uzoq muddatda xarajatlar ustidan nazorat saqlash uchun muammolarni yuzaga keltirishi mumkin.

Boshqa tomondan, Microsoft Azure monolitik platformasi biroz murakkabroq va sozlashi uchun ko'proq mehnat talab qiladi, ammo potentsial ravishda bardoshliroq va ancha katta miqdorda foydalanuvchilarga yetkazish imkonini beradi. Azure'ning asosiy afzalligi — uni barcha komponentlarni bir joyda birlashtirish imkoniyati. AI agentlar uchun Maxsus Yordamchalar, Container Apps, Functions, App Service kabi xizmatlar turlicha yuklamalarni qondirish uchun optimallashtirilgan (214, 222). Masalan, Telegram bot'ning webhook'larini Cloudflare Workers orqali qabul qilish 66 , 67 yoki render.com'da bot'ni "Always On" rejimida ishlatish 122 muammosini hal qilish uchun alternativ yechim bo'lishi mumkin. Bundan tashqari, Azure AI Foundry platformasi (125, 110) Gemini kabi tashqi modellar bilan ishlashni osonlashtiradi va agentni boshqarish, monitoring qilish va scale qilish uchun maxsus vositalarni taklif qiladi. Bundan tashqari, Azure'ning Global Presence va mahsulotlar aksariyati (masalan, Cosmos DB, Virtual Machines) 2026-yilda ham rivojlanayotgan ekotizimni qo'llab turadi (112).

Shunga qaramay, Azure'ning murakkabligi va narxlari (masalan, pay-as-you-go pricing modeli 146) ba'zi loyihalar uchun qimmatga tushishi mumkin. Migrantlarning narxlari turli xil bo'lishi mumkin, lekin umuman olganda, Azure'nin narxlari biroz murakkabroq va nazorat qilish qiyinroq bo'lishi mumkin. Shuning uchun, "Supabase+Render+Cloudflare" ekotizimi yuqori darajadagi skalirlash va maxsus infrastrukturaga ehtiyoj bo'lmaganda, tez start uchun yaxshi tanlov bo'lishi mumkin. Ammo, loyiha uzoq muddatda rivojlanishi, AI xizmatlariga chuqur integratsiya qilishi va ancha katta miqdorda foydalanuvchilarga yetkazishni nazarda tutsa, Azure'ning jami platformaviy yondashuvi va ancha katta resurslar bazosi unga o'zlashtirilishi mumkin bo'lgan murakkab mutaxassislikni talab qiladi.

Vendor Lock-in Risklari va Minimalizatsiya Strategiyalari

Har qanday cloud platformasiga tayanish bilan birga vendor lock-in riski doim mavjuddir. Ushbu qo'llanmada ko'rib chiqilgan arxitektura uchun bu riskni minimal darajada pasaytirish uchun bir nechta strategiyalar mavjud.

- 1. Interfeyslarni Standartlashtirish:** Har qanday platformadan foydalanishda ularning o'z interfeyslariga (API, SDK) tayanishdan voz kechish yoki ularni ancha oddiy wrapper'lar orqali qamrab olish muhim. Masalan, Gemini API'ga murojaat qilish uchun Google'ning rasmiy SDK'laridan emas, balki uni qo'llab quvvatlaydigan universal HTTP kliyentlardan foydalanish. Bu holda, agar Google'ning API siqilishi yoki nomlanishi o'zgarsa, API'ga murojaat qiladigan kodni o'zgartirish kerak bo'lganda faqat wrapper'ning ichki qismi o'zgartiriladi. Shunga o'xshash, Cloudflare R2'ning S3-kompatibil API'laridan foydalanish (55) R2'ni AWS S3 kabi boshqa object storage xizmatlariga o'tishni osonlashtiradi.
- 2. Containerizatsiya va Platformaneutral Deployments:** Docker konteynerlarini ishlatish (86) loyihani ma'lum bir hosting platformasidan mustaqil qilishning eng samarali usulidir. Eslatma: Dockerfile'ni optimallashtirish, multi-stage build'lar ishlatish va non-root foydalanuvchini sozlash kabi best practices'lar qo'llanilishi kerak (154). Render Container Apps (115) va Azure Container Apps (214) kabi xizmatlar Docker konteynerlarini host qilish uchun mo'ljallangan. Agar loyiha Docker konteyneri sifatida paketlanib, uning tarixi GitHub Container Registry'da saqlansa (213), uni istalgan joyga (masalan, AWS, GCP yoki o'z serverlariga) deploy qilish mumkin. Bu, ayniqsa, Render'ning "sleep" algoritmi (121) yoki boshqa platformaning cheklovlariga tegish muammosini oldini olishga yordam beradi.
- 3. Ma'lumotlar Bazasini Integratsiya uchun Maksimal Darajada Standard SQL Ishtirok Etish:** Supabase PostgreSQL'ni ishlatish, ma'lumotlar bazasi uchun standard Postgres imkoniyatlaridan foydalanishni talab qiladi. RLS siyosatlarini, indexlarni, transaktsiyalarni va shunga o'xshash barcha asosiy tushunchalarni standard Postgres sintaksisi bilan yozish. Bu, agar kelajakda ma'lumotlar bazasini Supabase'dan PostgreSQL on Azure (166) yoki boshqa hosting xizmatiga ko'chirish kerak bo'lsa, ma'lumotlar bazasi logikasini ko'chirishni osonlashtiradi. Ba'zi superkritik funksiyalar uchun Supabase'ning maxsus xizmatlaridan (masalan, Auth, Realtime) foydalanish zarur bo'lsa, ularning o'rnini bosuvchi boshqa xizmatlarni (masalan, Firebase Auth, Pusher) loyihaning arxitekturasiga qo'shishni rejalashtirish maqsaddir.

4. **Infrastructure as Code (IaC):** Loyihaning butun infratuzilmasini (frontend, backend, database schema, security groups) IaC vositalari (masalan, Terraform yoki ARM shablonlari) orqali boshqarish. Bu yerda har bir komponent uchun alohida vositalar mavjud: Render CLI (202), Supabase CLI (99), Terraform Cloudflare provider'i (206). IaC orqali infratuzilmani tasvirlash, uning versiyasini kuzatib boring va uni bir platformadan ikkinchisiga o'tkazish jarayonini avtomatlashtirish mumkin. Bu, ayniqsa, Disaster Recovery replikasini yaratish va uchinchi xosting provayderiga migratsiya qilishda juda foydalidir.

Quyidagi jadvalda "Supabase+Render+Cloudflare" va "Azure" ekotizimlari taqqoslanadi:

Texnologiya	Supabase+Render+Cloudflare Ekotizimi	Azure Monolitik Platformasi
Arxitektura Murakkablighi	O'rta (Turli xil xizmatlar o'rtasida integratsiya kerak)	Yuqori (Platforma keng, lekin murakkab)
Tez Start Imkoniyati	Yuqori (Git-to-cloud, minimal konfiguratsiya)	O'rta (Sozlash va integratsiya uchun ko'proq vaqt talab qiladi)
Vendor Lock-in Riski	O'rta (S3, Docker, standard SQL ni qo'llash orqali pasaytiriladi)	Yuqori (Maxsus xizmatlar va platforma integral yondashuvi)
Scalability	O'rta (Cheklovlar bilan bog'liq, masalan, bog'lanishlar)	Yuqori (Ancha katta resurslar bazasi, maxsus scaling strategiyalari mavjud)
AI/ML Imkoniyatlari	Gemini API kabi tashqi xizmatlardan foydalanish	Kuchli (Azure AI Foundry, Cognitive Services, OpenAI integration)
Narxlar	O'rta (Free tier'lar cheklangan, Pro planlar narxli)	O'rta/Yuqori (Pay-as-you-go, Enterprise tarmoqlari mavjud)
Security	RLS, IAM, JWT, OIDC	Active Directory, Key Vault, RBAC, Defender for Cloud
Monitoring	Grafana Cloud, Log Drains, Render Metrics	Application Insights, Log Analytics, Azure Monitor

Arxitektura Qarorlari (ADR)

Ushbu tadqiqot asosida quyidagi arxitektura qarorlarini (ADR) taqdim etish mumkin.

ADR-001: Asosiy Backend va Database Tanlovi

- **Context:** Loyihaning asosiy vazifasi Telegram botlari va mini-app'lar orqali foydalanuvchilar bilan AI agentlar orqali interaktivlikni ta'minlash. Bu, tez so'rov-qabul javoblari, ma'lumotlarni ishlov chiquvchi, autentifikatsiya va real-time xabarlar talab qiladi.
- **Options Considered:**
- **A: Supabase + Render:** Supabase'ning PostgreSQL'ni backend sifatida, Render'da Node.js/Python veb-serverini ishga tushirish.

- **B: Azure App Service + Azure Database for PostgreSQL:** Ikkala komponentni ham Microsoft platformasida host qilish.
- **C: Cloudflare Workers + D1:** Barcha compute va storage'ni Cloudflare platformasida birlashtirish.
- **Decision: A tanlandi.** Supabase'ning RLS, Auth va Realtime kabi xizmatlari (274) render.com bilan integratsiyasi ancha oson va tezdir (48). Bu yondashuv loyihani ularning API'larini ishlatish orqali alohida sozlash imkonini beradi.
- **Why:** Tez start, minimal konfiguratsiya va minimal operatsion xarajatlar (DevEx). Supabase'ning RLS mexanizmi ma'lumotlar bazasi darajasida himoya qilish uchun juda samarali.
- **Consequences:**
- **Positive:** Jamoaning dasturlashga va foydalanishga qaratilishi oson. Bepul rejalardan foydalanish deyarli xarajatsiz sinov uchun yaroqli.
- **Negative:** Vendor lock-in riski. Narxlar free rejaning cheklovlaridan keyin keskin oshishi. Bog'lanishlar pool exhaustion kabi muammolar yuzaga kelishi mumkin (90).

ADR-002: AI Model Routing Strategiyasi

- **Context:** Loyiha turli xil murakkablikdagi AI so'rovlarini qabul qiladi: oddiy matnli so'rovlar, murakkab tahlillar, tasvir yaratish va video generatsiyasi. Har bir model turining xarajati va tezligi farq qiladi.
- **Options Considered:**
- **A: Single Model Approach:** Faqat eng kuchli (va qimmat) modeldan (masalan, Gemini 3.5 Pro) foydalanish.
- **B: Rule-Based Routing:** So'rovning turiga qarab turli modellarni tanlash (masalan, Flash uchun oddiy so'rovlar, Pro uchun murakkablar).
- **C: Dynamic Routing with LLM:** Birinchi LLM so'rovini jo'natib, u so'rovning murakkabligini baholay va to'g'ri modelni tanlaydi.
- **Decision: B tanlandi.** Rule-based routing eng samarali va nazorat qilinadigan yondashuv.
- **Why:** So'rovlar xarajatlarini maksimal darajada kamaytiradi. Masalan, `gemini-3.8-flash` modeli odatda `gemini-3.5-pro` dan 10 baravar arzon (6, 228). Bu strategiya tezkorlik va xarajat optimallashtirish uchun ideal.
- **Consequences:**
- **Positive:** Xarajatlar nazorat ostida. Javoblar tezroq.
- **Negative:** So'rov turini aniqlash uchun qo'shimcha kod yozish talab etiladi. Aniq klassifikatsiya qilinmagan so'rovlar uchun hech qanday optimallashtirish amalga oshirilmaydi.

ADR-003: Data Persistence and Storage Strategy

- **Context:** Loyiha turli xil ma'lumotlarni saqlashni talab qiladi: foydalanuvchi profillari, chat tarixi, AI agent state'lari va media fayllari (tasvir, video).
- **Options Considered:**
- **A: Supabase Storage:** Barcha ma'lumotlarni PostgreSQL (profil, tarix) va Supabase'ning ob'ektlilik xizmati (media) da saqlash.
- **B: Cloudflare R2:** Barcha ma'lumotlarni PostgreSQL (profil, tarix) va R2'da (media) saqlash.
- **C: Azure Blob Storage:** Barcha ma'lumotlarni PostgreSQL (profil, tarix) va Azure Blob Storage'da (media) saqlash.
- **Decision: B tanlandi.** R2'ning "zero egress fee" modeli (160) uning eng katta afzalligidir. Supabase Storage'ning meta ma'lumotlari PostgreSQL'ga bog'langanligi sababli, R2'ning alohida ob'ektlilik bazasi bilan ishlash o'ziga xos imkoniyatlarni beradi.
- **Why:** Saqlash xarajatlarini minimal darajada saqlash. R2'ning global CDN'ga integratsiyasi yuklash tezligini oshiradi.
- **Consequences:**
- **Positive:** Media fayllari yuklanishi va yuklab olinishi uchun xarajatlar yo'q. Global tezlik.
- **Negative:** R2'ning API'ga murojaat qilish uchun qo'shimcha kod yozish kerak. R2'ning free rejasi cheklovlariga e'tibor berish kerak (10 GB storage, 1M Class A va 10M Class B requests) (163).

Ushbu arxitektura qarorlari loyihaning 100 dan 100,000 gacha foydalanuvchilarga yetkazishini ta'minlaydi. Har bir qaror tanlangan texnologiyalar va ularning 2026-yil holatidagi cheklovlariga asoslangan. Ushbu qo'llanmaning keyingi bo'limlari har bir tanlangan texnologiya uchun chuqur tahlilni takomillashtiradi.

Supabase: Backend-as-a-Service va Database Management

Supabase 2026-yil holatida PostgreSQL asosida qurilgan, tez-tez yangilanib borayotgan va AI xizmatlariga moslashtirilgan, tejamkor Backend-as-a-Service (BaaS) platformasi sifatida namoyon bo'lyapti. Uning asosiy tarkibiy qismlari PostgreSQL ma'lumotlar bazasi, Supabase Auth, Storage, Realtime, Edge Functions va REST API gateway'ni o'z ichiga

oladi. Ushbu bo'limda Supabase'ning arxitekturası, xavfsızlık mexanizmlari, debugging strategiyalari va 2026-yil holatidagi bepul reja cheklovlariga chuqur tahlil qilinadi.

Supabase Arxitekturasi va Ishlash Prinsiplari

Supabase'ning arxitekturasini PostgreSQL'ning kuchliqtirilgan imkoniyatlariga asoslangan. Uning asosiy komponentlari quyidagilardan iborat:

- **PostgreSQL:** Har bir projekt uchun alohida alohida ma'lumotlar bazasi ajratiladi. Supabase'ning freemium rejimida har bir projekt uchun 500 MB'lik hajm cheklovi mavjud (26 , 23). Bu cheklov 2026-yil mart oyida har bir projekt uchun alohida o'lchovga o'tkazildi, bu esa birdan-birdan ko'payayotgan loyihalar uchun yanada nozikroq tartibga solish imkonini beradi (182).
- **Supabase Auth:** Avtomatlashtirilgan, passkey'lar kabi yangi autentifikatsiya usullarini qo'llab quvvatlaydigan, tayyorlashga tayyor avtorizatsiya tizimi (265). U Telegram autentifikatsiyasi kabi tashqi provayderlar bilan integratsiya qilish imkonini beradi.
- **Storage:** Supabase'ning ob'ektlilik xizmati, uning API'si S3-kompatibil (55). U PostgreSQL'dagi meta ma'lumotlar bilan bog'langan va RLS orqali himoyalangan.
- **Realtime:** PostgreSQL'ning LISTEN/NOTIFY mexanizmidan foydalanib, ma'lumotlar bazasidagi o'zgarishlarni klientga aniq bir vaqtga yaqin bayon qilish imkonini beradi. Bu, chat ilovalari va real-time dashboard'lar uchun juda foydali (274).
- **Edge Functions:** Serverless funksiyalar, ularni JavaScript (Node.js) yoki Python kabi tillarda yozish imkonini beradi. Ular Supabase'ning barcha xizmatlari bilan birgalikda ishlay oladi (58).
- **REST API Gateway (PostgREST):** Har bir PostgreSQL jadvallari uchun avtomatlashtirilgan REST API endpoint'larini yaratadi. Bu, backend'ning oddiy qismini olib tashlashga imkon beradi.

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bog'lanishlar..... **Deconstruction of the User's Need:**

1. **Asosiy Maqsad:** 2026-yil 6-sentyabr holatiga mos keladigan, Telegram Mini App, Bot va AI agentni birlashtiruvchi hybrid arxitekturani yaratish uchun senior dasturchilar uchun amaliy qo'llanma ishlab chiqish.
2. **Texnologiya Stakki:** Supabase, Render, GitHub, Cloudflare (Pages/R2), Telegram Platform, Gemini API, Azure.
3. **Kritik Talablar:**
 - **Vaqt Chegarasi:** Barcha ma'lumotlar 2026-yil 6-sentyabr holatiga mos kelishi shart.
 - **Yangi Ma'lumotlar:** Aynan 2026-yil 1-6-sentyabr oralig'idagi yangilanishlar alohida ajratilishi kerak.
 - **Amaliylik:** Qo'llanma faqat nazariya emas, balki "senior developer production handbook" sifatida amaliy maslahatlarni o'z ichiga olishi lozim.

1. Tadqiqotning Asosiy Yo‘nalishlari:

- **Arxitektura:** Har bir texnologiya uchun optimal arxitektura tanlovlari va arxitekturaviy savdo-sotdiqlar (ADR).
- **Xavfsizlik:** JWT, RLS, webhook security, secret management kabi muhim sohalarni chuqur tahlil qilish.
- **Bepul Rejim Strategiyasi:** Har bir komponentning bepul rejimini minimal xarajat bilan maksimal foydalanuvchiga yetkazish uchun optimallashtirish.
- **Xarajatlar va Masshtablash:** Foydalanuvchi soni (100 dan 100,000) o'sishi bilan resurs sarfi (tokenlar, DB so'rovlari, saqlash) bo'yicha hisob-kitoblarni amalda tekshirish.
- **Muhim Ssenariylar:** Webhook'lar, long-polling, cold start, rate limits, connection exhaustion, prompt caching kabi muammolarni hal qilish bo'yicha diagnostika yo'riqnoma va debugging daraxtlari.
- **CI/CD va DevOps:** GitHub Actions orqali to'liq CI/CD jarayonini avtomatlashtirish, deploy, monitoring, rollback strategiyalarini ishlab chiqish.

- **Vendor Lock-in:** Har bir texnologiyadan foydalanishning oqligi va oqibatlarini baholash.

1. **Yakuniy Deliverable:** Ushbu tadqiqot natijasida bitta yakuniy qo'llanma mavjud bo'lishi kerak bo'lgan "Senior Engineer Knowledge Map", "Production Checklist", "Master Cheat Sheet" va "Final Recommended Stack" kabi aniq materiallar ham talab etiladi.

Report Outline:

- **Level 1 Title:** 2026 Yili uchun Hybrid Telegram AI Agent Arxitekturasini Senior Dasturchilar uchun Amaliy Qo'llanma
- **Level 2 Titles:**
 - 2026-Yil 6-Sentyabr Holatidagi Asosiy Texnologiyalar va Yangilanishlar
 - Supabase: Backend, Auth va Storage Arxitekturasini
 - Render va Azure: Compute, Hosting va AI Infrastrukturasini
 - Cloudflare: Frontend, CDN va Object Storage Arxitekturasini
 - Telegram Bot va Mini Apps: Integratsiya, Xavfsizlik va Erkinliklar
 - Gemini API: AI Agent Arxitekturasini, Token Optimallashtirish va Narxlantirish
 - Yakuniy Arxitektura, Xarajat Boshqaruvi va Mas'uliyatli Dasturlash Praktikasi

Hybrid Telegram AI Agent Arxitekturasini: 2026-yilda 0dan 50 gacha xarajatlarni bilan Productionga Chiqish G'ildiragi

Ushbu tadqiqot, 2026-yil sentyabr holatiga mos keladigan, Telegram Mini App, Telegram Bot va AI agentni birlashtiruvchi murakkab arxitekturaning yaratish, boshqarish va nazorat qilish bo'yicha amaliy qo'llanma ishlab chiqishga qaratilgan. Tadqiqotning asosiy maqsadi — senior dasturchilar uchun maxsus mo'ljallangan, zamonaviy va amalga oshirishga tayyor texnik bilimlarni yagona manzilga yig'ish. Ushbu qo'llanma har bir asosiy texnologiya uchun arxitekturaviy savdo-sotdiqlar, xavfsizlik standartlari, xarajat optimallashtirish strategiyalari va masshtablanish chegaralari haqida chuqur tushuncha beradi. Barcha ma'lumotlar 2026-yil 1-6-sentyabr oralig'idagi rasmiy changelogs, dokumentatsiyalar va narxlar ro'yxati asosida tekshirilgan va tasdiqlangan bo'lib, eski yoki noto'g'ri ma'lumotlardan xoli qilingan. Loyihaning konteksti — hybrid model bo'lib, Telegram Bot oddiy interfeys va tezkor vazifalar uchun, Mini App esa murakkab AI interfeyslari, media yuklash va boy funksionallik uchun ishlatiladi. Bu modelda bot va

mini app o'rtasida umumiy foydalanuvchi identifikatori, chat history va backend logikasi mavjud bo'ladi. Tadqiqot 100 dan 100,000 gacha foydalanuvchilarga mo'ljallangan loyihalar uchun resurs sarfini (AI tokenlari, DB so'rovlari, saqlash operatsiyalari, webhook yuklanishi) Low/Medium/Heavy ssenariylari bo'yicha tahlil qiladi. Yakuniy natijada, deyarli xarajatsiz (0), *minimal*(5), o'rta (25) *vato'liq masshtab* *lantirilgan* (50+) variantlardagi arxitekturalar taklif etiladi. Shuningdek, loyiha deployidan oldin tekshirish ro'yxati, debugging yo'riqnoma va senior mutaxassislar uchun bilim xaritasi kabi amaliy vositalar ham jamlanadi. Ushbu qo'llanma faqat texnologiyalarni qanday ishlatishni emas, balki nima uchun aynan shu texnologiya tanlanganligi, qaysi nuqtada boshqa variantga o'tish kerakligi va uzoq muddatli oqibatlari haqida chuqur tushuncha beradi.

2026-Yil 6-Sentyabr Holatidagi Asosiy Texnologiyalar va Yangilanishlar

2026-yil holatini tahlil qilishdan oldin, ushbu tadqiqot uchun asos bo'layotgan texnologiyalar ekotizimining umumiy holatini va keyingi bosqichlarda chuqurroq tahlil qilinadigan muhim yangilanishlarni tushuntirish zarur. Ushbu qismda har bir platforma uchun asosiy imkoniyatlari, narxlari va 2026-yil avgust oyida sodir bo'lgan o'zgarishlar bayon etiladi. Taqdim etilgan manbalardan ko'rinib turibdiki, 2026-yilning yozida asosiy platformalar o'zlarining arxitekturalari va narx politikalarini ustidan qo'yishga davom etmoqda, bu esa yangi yechimlar va optimallashtirish imkoniyatlarini ochadi.

Supabase 2026-yilgi holatida Postgres development platformasi sifatida o'zini namoyish etmoqda ⁶¹. Platformaning freemium modeli asosan o'zgarmagan, lekin ba'zi muqobil yechimlar oqibatlarini yoritib beradi. Supabase'ning Free plani hozirda 500 MB ma'lumotlar bazasi hajmi, 50,000 MAU (oylik faol foydalanuvchi), 5 GB egress bandwidth va 1 GB file storage ni o'z ichiga oladi ^{23 26 165}. Eng muhim o'zgarishlardan biri — 500 MB'lik ma'lumotlar bazasi limiti hozirda har bir projekt uchun alohida belgilanganligidir, bu esa bir organizatsiya ichida bir nechta proyektlar yaratish imkonini beradi ¹⁸². Pro plani esa \$25/ora bilan boshlanadi ¹²⁰. Changelog'da 2026-yil may oyida yangi SDK'ning e'lon qilinishi va noyabr oyida Edge Functions'ga AI model qo'llanilishi haqida ma'lumotlar boradi ^{55 56}. Bundan tashqari, Supabase MCP serverini va agent skills'ni integratsiya qilish imkoniyati tavsiya etilmoqda, bu esa AI agent'lar uchun yanada chuqur integratsiya yaratadi ¹⁸⁵.

Render platformasi ham 2026-yilda o'zining narx siyosatini qayta ko'rib chiqqan. Avgust 1, 2026 dan boshlab, Seat fees olib tashlandi va Workspace plans o'zgartirildi ¹¹⁵. Hozirda Render'ning asosiy rejalari Hobby (eski nomi), Pro (25/ora) va Scale (499/ora) planlarini o'z ichiga oladi ¹¹⁸. Hobby rejasida 750 ta bepul instance soatlari, 5 GB egress bandwidth va 30 kunlik PostgreSQL ma'lumotlar bazasi saqlanmasi mavjud ¹¹⁴¹⁹². Lekin bu rejimda 15 daqiqa iste'mol mavjudligi va 5 GB egress limiti kabi cheklovlar mavjud ¹²¹. Pro reja esa \$25/ora bilan boshlanib, unga qo'shimcha resurslar (CPU, RAM) uchun pullik xizmatlar qo'shiladi ¹¹⁷. Platformaning changelog'i render.com saytidagi o'zgarishlarni o'z ichiga oladi ¹⁹³²⁰³. Bundan tashqari, Render Workflows'ning lanchi agent logic, data pipelines va billing flows kabi uzoq muddatli jarayonlar uchun ishlatilishi mumkin ²²⁰.

Cloudflare platformasi hozirda ham unikal narx modeli bilan ajralib turadi, ayniqsa Cloudflare R2 object storage xizmatida. R2'ning eng katta afzalligi — uni egress bandwidth'ining haddan tashqari yuqori bo'lishi bilan bog'liq. Bu, AWS S3 kabi xizmatlardan farqli ravishda, ma'lumotlarni yuklab olish uchun hech qanday to'lov qilmaslikni ta'minlaydi ¹⁶⁰¹⁶¹. R2'ning free tier'i 10 GB storage, 1 million Class A va 10 million Class B requests'ni o'z ichiga oladi ¹⁶³. September 1, 2026 dan boshlab, Workers Free plan'dagi D1'ning row read/write limitlariga tegish bo'ldi va query'lar shunga bog'liq bo'lib qoldi ⁹⁸. D1'ning free plani esa 5 GB hajmdagi ma'lumotlarni saqlash imkonini beradi ¹⁹⁸. Cloudflare Pages'ning free rejasida 500 ta deploy'lar o'rtacha sanaladi ⁴⁴.

Gemini API ekotizimi 2026-yilda yanada murakkabroq va model turlariga bo'lingan. Google, turli xizmatlar uchun turli modellar taklif etadi. gemini-3.8-flash modeli, 1,048,576 token'li context window'ga ega bo'lgan tezkor va arzon flash model sifatida e'lon qilingan ¹⁵⁰. gemini-omni-1.1-flash esa video generatsiyasi va editing uchun mo'ljallangan ²²⁹, gemini-3.5-pro esa murakkab tahlil va kod yozish uchun mo'ljallangan ²²⁸. Model narxlari taxminan 0.10–0.40 (input/output) atrofida bo'lib, gemini-2.5-flash-lite modelining 0.10/0.40 narxi misol bo'la oladi ⁷⁴. Gemini API'ning rate limits ham mavjud va ular turli usullar va tizimlar uchun farq qiladi ^{72 253}. So'rovlar chegarasini oshirish uchun Google AI Studio'da so'rov qo'shish mumkin ²²⁴.

Azure platformasi 2026-yilda rivojlanayotgan ekotizimni qo'llab turadi, ayniqsa Microsoft Foundry platformasi. Foundry agent service — bu AI agentlarni yaratish, deploy qilish va monitoring qilish uchun mo'ljallangan boshqaruv platformasi ¹²⁵²²². Bu, ayniqsa, korporativ sifatida ishlash uchun mo'ljallangan bo'lib, agent orchestration, observability va secure model operations kabi xizmatlarni taklif etadi ¹¹⁰. Azure'ning AI xizmatlari, ayniqsa, Gemini API bilan integratsiya qilish uchun muhim alternativ bo'lib xizmat qiladi

136. Narxlar murakkabroq bo'lsa-da, Enterprise sifatida ishlash uchun mo'ljallangan. Azure'ning AI Foundry vs. Gemini Enterprise taqqoslamasi 2026-yilda yanada aniq bo'lib, har birining o'z afzalliklari mavjud 145223.

Telegram platformasi 2026-yilning yozida o'z API'si va platforma imkoniyatlari bo'yicha sezilarli o'zgarishlarni e'lon qilmagan. Taqdim etilgan changelogs'lar (NVidia Omniverse, FF14, NARAKA: BLADEPOINT kabi o'yinlar) Telegram'ning o'z platformasining yangilanishlaridan emas, balki boshqa kompaniyalar o'z platformalarida qilgan o'zgarishlarni aytib beradi 275277282. Demak, 2026-yil sentyabr holatida Telegram Bot API va Mini Apps'lar uchun asosiy imkoniyatlar va cheklovlar 2025-yilgi holatiga o'xshash qoldi. Bu esa, arxitekturaviy qarorlar qabul qilishda stabil va prognoz qilish mumkin bo'lgan platforma imkoniyatlaridan foydalanish imkonini beradi.

Platforma	Asosiy O'zgarishlar / Afzalliklar	Free Tier Limitlari (2026-yil sentyabr holatida)
Supabase	Pro plan \$25/ora; har bir projekt uchun alohida 500MB DB limiti 182; MCP server integratsiyasi 185.	500 MB DB, 1 GB Storage, 5 GB Egress, 50,000 MAU 23 26 .
Render	Narxlar o'zgargan (Pro \$25/ora); Hobby rejasida 750 soat, 5 GB egress 114; Workflows uchun maxsus reja 220.	750 soat, 5 GB egress, 30 kunlik DB 192.
Cloudflare	R2 zero egress fee modeli; D1'da row limitlari oshirildi (Sep 2026) 98 .	10 GB Storage, 1M Class A, 10M Class B R2 requests 163.
Gemini API	Turli xizmatlar uchun turli modellar (Flash, Pro, Omni) mavjud 150229; narxlar aniqlashgan.	Aniq raqamli free tier mavjud emas, ammo rate limits mavjud 6 .
Azure	Microsoft Foundry agent service — AI agentlar uchun maxsus platforma 125.	Free credits (\$200) va bir nechta xizmatlarning bepul rejimlari mavjud 109.

Shunday qilib, 2026-yil holatida texnologiyalar ekotizimi o'zgarishsiz qolmagan. Supabase va Render freemium modelini saqlab, lekin narxlar va cheklovlarini moslashtirmoqda. Cloudflare R2'ning zero egress modeli hozirda ham unikal afzallikdir. Gemini API ekotizimi yanada tartibliroq bo'lib, turli xizmatlar uchun turli modellar va narxlarni taklif etadi. Azure esa korporativ va rivojlangan AI agent loyihalari uchun eng kuchli alternativ bo'lib qolmoqda. Ushbu yangilanishlar va o'zgarishlar keyingi boblarda har bir texnologiya uchun arxitekturaviy qarorlarni qabul qilishda asos bo'lib xizmat qiladi.

Supabase: Backend, Auth va Storage Arxitekturası

Supabase 2026-yilda Postgres development platformasi sifatida o'zini namoyish etmoqda va loyihalarni tezda yaratish va boshqarish uchun kuchli vosita bo'lib qolmoqda ⁶¹.

Ushbu bo'limda Supabase'ning asosiy komponentlari — PostgreSQL, Supabase Auth, Storage, Realtime va Edge Functions — ularning arxitekturası, xavfsızlik mexanizmlari, bepul rejim cheklovlari va muammolarni hal qilish strategiyalari chuqur tahlil qilinadi. Ushbu tahlil senior dasturchilar uchun Supabase'ni prodakshen muhitida samarali va xavfsız tarzda qanday qurish, boshqarish va optimallashtirish mumkinligini ko'rsatadi.

Supabase'ning arxitekturası PostgreSQL ma'lumotlar bazasiga asoslangan bo'lib, undan tashqari Supabase Auth, Storage, Realtime va Edge Functions kabi xizmatlarni qo'shib beradi. PostgREST, Supabase'ning REST API yaratish uchun ishlatadigan vositasidir, bu esa SQL so'rovlari yozmasdan ma'lumotlar bazasi ustida ishlash imkonini beradi ¹⁰⁸.

Supabase'ning xavfsızlik arxitekturası ikkita asosiy kalitdan iborat: Row Level Security (RLS) va xususiyatlar. RLS, har bir foydalanuvchi uchun ma'lumotlar bazasidagi qatorlarni himoya qilish uchun mo'ljallangan. Supabase'ning RLS siyosatlarini to'g'ri sozlash — bu ma'lumotlarni buzishdan himoya qilishning eng muhim bosqichi hisoblanadi ³⁴. Supabase'ning har bir proyektda ikkita kalit mavjud: `anon` kaliti va `service_role` kaliti. `anon` kaliti frontend ilovalarida foydalanilishi uchun mo'ljallangan va RLS siyosatlariga rioya qiladi. `service_role` kaliti esa backend ilovalarida (masalan, Edge Functions) foydalaniladi va RLS siyosatlarini bekor qiladi, bu esa superuser huquqlarini beradi. Shuning uchun `service_role` kalitini hech qachon frontend yoki `открыто` joylashtirilgan joylarda saqlash mumkin emas ³⁴.

Supabase'ning bepul rejim cheklovlari 2026-yil holatida quyidagilarni o'z ichiga oladi: har bir projekt uchun 500 MB ma'lumotlar bazasi hajmi, 1 GB fayl saqlash, 5 GB egress bandwidth (oqimdan chiqish) va oyiga 50,000 MAU (oylik faol foydalanuvchi) ^{23 26 165}.

Eng katta xavfli nuqta bu rejimda projekt bir hafta davomida faol bo'lmasa, u avtomatik ravishda to'xtatiladi va qayta yoqilishi uchun qayta tiklash bozorida pul to'lash talab qilinadi ¹⁶⁴. Agar ikkita yoki ko'proq projektlar uchun bepul rejimga kirib ketilsa, har bir projekt uchun alohida to'lov qilish talab etiladi, ya'ni bir projekt uchun

25 to'lashning natijasi ikkinchi projekt uchun ham xuddi shunday to'lovni talab qiladi ^{[[168]]}. Supabase'ning 2 hisoblanadi va har bir rejadagi bepul cheklovlar bu xizmatdan foydalanishni cheklashi mumkin ⁵⁹.

Supabase'ning eng keng tarqalgan muammolari ma'lumotlar bazasi bog'lanishlari bilan bog'liq. "Database connection exhausted" (Ma'lumotlar bazasi bog'lanishi tugagan) xatosi, ayniqsa, yuklama oshganda paydo bo'ladi. Bu muammo, agar veb-server har bir so'rov

uchun yangi bog'lanish ochib, uning ishini tugatgach, unga yopishini boshlaysa, yuzaga keladi ⁹⁰. To'g'ri yondashuv — bu bog'lanishlarni oqimida saqlash va ulardan qayta foydalanishdir. Supabase'ning `Supervisor` deb nomlangan bog'lanishlar oqimida saqlash vositasi bu muammoni hal qilish uchun mo'ljallangan ⁸⁹. `Supervisor`'ni yoqish va portni 65000 portidan 6501 portiga o'zgartirish orqali bog'lanishlar sonini kamaytirish va boshqarish mumkin ⁸⁹. Shuningdek, RLS siyosatlarining noto'g'ri sozlanganligi ham muammolarni keltirib chiqarishi mumkin. Agar foydalanuvchi ma'lumotlarni o'qiy olmasa, RLS siyosati noto'g'ri sozlangan ekanligini anglatadi. Unified Logs funksiyasi yordamida (`Unified Logs is now in open beta`) barcha xizmatlardan (API gateway, Postgres, Auth, Storage) loglarni bitta joyda yig'ish mumkin bo'lib, bu diagnostika jarayonini osonlashtiradi ²⁶⁴.

Supabase'ning xavfsizlik mexanizmlari juda muhim. RLS'ni to'g'ri sozlash — bu eng muhim bosqichdir. Har bir jadvallar uchun RLS siyosatlarini yozish va ularni `authenticator` roliga bog'lash lozim. Fayllarni saqlash uchun `Storage policies` mavjud bo'lib, bu sizga har bir foydalanuvchiga faqat o'z fayllariga kirish imkonini beradi. Supabase'ning auth tizimi esa enterprise-managed auth kabi keng ko'lamli xizmatlarni qo'llab chiqadi ²⁴³. `Passkeys`'ning beta versiyasi ham e'lon qilingan bo'lib, bu esa parolga asoslangan autentifikatsiyadan xoli bo'lish va phishing hujumlariga chidamliroq autentifikatsiya usuli bo'lishi mumkin ²⁶⁵. Supabase'ning Log Drains funksiyasi Pro rejasida mavjud bo'lib, bu esa loglarni tashqi manzillarga (masalan, Splunk, Datadog) yuborish imkonini beradi ¹⁸⁸. Bu esa korporativ muhitda loglarni markazlashtirib boshqarish uchun muhimdir.

Supabase'ning real-time xizmati — bu Postgres ma'lumotlar bazasidagi o'zgarishlarni klientlar tomonga millisekundlar ichida jo'natish imkonini beruvchi kuchli funksiyadir ²⁷⁴. Agar Realtime xizmati ishlamasa, bunda quyidagilarni tekshirish kerak: 1. **PostgREST aktivligini tekshiring:** Realtime xizmati faoliyat yuritish uchun PostgREST (Data API) faol bo'lishi kerak. Agar Data API o'chirilgan bo'lsa, Realtime ham ishlamaydi ¹⁰⁸. 2. **Loglarni tekshiring:** Unified Logs yoki Log Drains orqali Realtime xizmatining o'ziga xos loglarini tekshiring ²⁶⁴. 3. **Supabase dashboard'da Realtime tab'ini tekshiring:** Dashboard'da Realtime xizmatining holatini ko'rish mumkin. Agar u o'chirilgan bo'lsa, uni yoqing. 4. **Network/Websocket muammosi:** Klient tomonidagi firewall yoki proxy'lar WebSocket ulanishlarini bloklab turishi mumkin. Bu holda, SSL/TLS sertifikatlarining to'g'riligini va HTTPS ulanishlarining to'g'riligini tekshirish kerak.

Supabase'ning Edge Functions'ga oid muammolar ham ko'p uchraydi. 504 Gateway Timeout xatosi, funksiya 150 sekunddan ortiq ishlashini anglatadi, bu esa funksiya uchun maksimal ishlash vaqtidir ¹⁰⁴. Agar Edge Function 504 xatosi bersa, funksiya

ichida infinite loop yoki juda uzun vaqt davom etadigan ishlar bajarilayotganini tekshirish kerak. Supavisor'ning "Circuit breaker open" xatosi esa, ma'lumotlar bazasi parolini almashtirgandan so'ng, funksiyaning yangi parol bilan bog'lanishga harakat qilishi mumkin ¹⁰⁵. Bu holda funksiyaning konfiguratsiyasini va parolni tekshirish kerak.

Supabase'ning xavfsizlik protokollari quyidagi jadvalda xulosalandi:

Muammo	Sabab	Diagnostika	Hal qilish	Prevenziya
RLS Blokiyapti	Noto'g'ri siyosat yoki ro'yxatdan o'tkazilmagan foydalanuvchi	authenticator ro'yxati va siyosatni tekshirish ³⁴	Siyosatni to'g'ri yozish va authenticator ro'yxati to'g'riligini tekshirish	RLS'ni har doim yoqish va siyoslarni test qilish ¹⁹⁰
API 401 (Unauthorized)	Noto'g'ri yoki eskirgan JWT	JWT'ni tokenni tekshirish va supabase.auth.getUser() orqali tekshirish	Refresh token orqali yangi access token olish	Access token'ni serverda saqlash va uning amal qilish muddatini nazorat qilish
API 403 (Forbidden)	RLS siyosatiga rioya qilmagan	Unified Logs orqali RLS siyosatiga bog'liq hatoliklarni izlash ²⁶⁴	RLS siyosatini to'g'ri sozlash	Har bir jadval uchun aniq RLS siyosatlarini yaratish
Connection Exhausted	Bog'lanishlar oqimida saqlash yoqilmagan yoki cheklovlar chekka yetdi	Supavisor holatini tekshirish va bog'lanishlar sonini nazorat qilish ⁸⁹	Supavisor'ni yoqing va bog'lanishlar oqimida saqlashni optimallashtiring	Veb-serverda bog'lanishlar oqimida saqlashni sozlash va cheklovlar sonini oshirish
Realtime Ishlamayapti	PostgREST o'chirilgan yoki network muammosi	PostgREST holatini tekshirish va websocket ulanishini sinovdan o'tkazish ¹⁰⁸	PostgREST'ni yoqing va network sozlamalarini tekshiring	Realtime'ni har doim yoqish va testing qilish

Umuman olganda, Supabase prodakshen muhitida yaxshi ishlashi uchun ma'lumotlar bazasi bog'lanishlarini to'g'ri boshqarish, RLS siyosatlarini chuqur tushunish va bepul rejim cheklovlarini hisobga olish muhimdir. Supavisor'ni yoqish, Edge Functions'lar uchun timeout chegaralarini to'g'rilash va Unified Logs'ni foydalanish kabi amaliy choralar muammolarni oldini olish va hal qilishda samarali bo'ladi.

Render va Azure: Hisoblash, Joylashtirish va AI Infrastrukturasini Tahlil Qilish

Backend hisoblash va AI infratuzilmasi arxitekturasini tanlovi — loyihaning skalirlanishi, xarajatlari va operatsional murakkabligi uchun asosiy muhimi hisoblanadi. Ushbu bo'limda Render va Azure platformalari turli xizmatlari (Web Services, Background

Workers, Functions, Container Apps) orqali 2026-yil holatiga mos keladigan arxitekturaviy qarorlar chuqur tahlil qilinadi. Taqqoslash mezonlari sifatida yuklama modeli, sozlash murakkabligi, xarajatlar, skalirlanish imkoniyatlari va vendor lock-in riski olinadi.

Render platformasi Node.js, Python kabi dasturlash tillarida yozilgan veb-ilovalarni joylashtirish uchun mo'ljallangan. Uning asosiy xizmatlari Render Web Service, Background Worker va PostgreSQL ma'lumotlar bazasi hisoblanadi. Render'ning web service'lari Docker konteynerlarini qo'llab, Git repozitoriyasiga push qilish orqali avtomatik deploy qilishni ta'minlaydi ¹⁴⁷. Lekin Telegram bot'lar uchun Render'ning "sleep" funksiyasi — bu muammolarni keltirib chiqaradigan muhim cheklov hisoblanadi. Agar bot 15 daqiqa davomida hech qanday so'rov qabul qilmasa, Render konteynerini to'xtatadi ¹²¹¹²². Birinchi so'rov kelgach, konteyner qayta ishga tushadi va bu "cold start" jarayoni bir necha soniyadan bir necha daqiqagacha davom etishi mumkin, bu esa Telegram bot'larining real vaqt rejasida javob berish talablariga zid. Shuning uchun Telegram bot'larini joylashtirish uchun Render'da "Always On" rejasini sozlash yoki bot'ni Cloudflare Workers kabi boshqa xizmatga ko'chirish tavsiya etiladi ⁶⁷. Render'ning background workers ham veb-service'lar kabi ishlaydi va sozlash oson, lekin ularni sozlash uchun veb-service'ni sozlashga o'xshash narxlarni to'lash kerak bo'lishi mumkin ¹¹⁹.

Azure platformasi esa biroz murakkabroq, lekin potentsial ravishda bardoshliroq va ancha katta miqdorda foydalanuvchilarga yetkazish imkonini beradi. Azure App Service, Azure Functions, Azure Container Apps kabi xizmatlar mavjud. Azure App Service — bu traditional veb-hosting xizmati bo'lib, Render'ning web services'iga yaqin. Lekin u container'lar bilan ishlashga mo'ljallangan. Azure Functions esa serverless xizmat bo'lib, Telegram webhook'larini qabul qilish uchun ideal. Ular chiroyli skalirlanadi va faqat so'rovlar vaqtida to'lanadi. Shuningdek, Azure Functions'lar Python, Node.js kabi tillarda yozilishi mumkin ²¹¹. Azure Container Apps esa Kubernetes'ga yaqin, lekin unchalik murakkab emas. Ular container'lar oqimini boshqarish, scaling va networking'ni serverless darajasida ta'minlaydi. Ushbu xizmatlarni tanlashda yuklama modeli muhim rol o'ynaydi. Agar yuklama o'zgaruvchan bo'lsa (masalan, Instagram kabi), serverless xizmatlar (Azure Functions) xarajatlar uchun samaraliroq bo'ladi. Agar yuklama barqaror bo'lsa, container'lar (Azure Container Apps) yoki traditional hosting (Azure App Service) yuqori performansni ta'minlaydi.

Taqqoslash jadvali quyidagicha:

Xizmat	Render Web Service	Azure App Service	Azure Functions	Azure Container Apps
Yuklama Modeli	Barqaror, o'z-o'zidan skalirlanmaydi	Barqaror, vertikal va gorizontal skalirlanadi	O'zgaruvchan, serverless	O'zgaruvchan, serverless
Sozlash Murakkabligi	Oson	O'rta	O'rta	Murakkab
Cold Start	Mavjud (agar sleep bo'lsa)	Mavjud	Mavjud (birinchi so'rov uchun)	Mavjud
Docker Support	Ha	Ha	Ha	Ha
Background Jobs	Background Worker	Logic Apps yoki Timer trigger	Timer trigger	Standart
Xarajatlar	Pro (\$25/ora + compute) 117	Pay-as-you-go (compute, storage)	Pay-per-execution	Pay-per-second (CPU/RAM)
Vendor Lock-in	O'rta	Yuqori	O'rta	O'rta

AI agent arxitekturasini uchun Azure'ning AI xizmatlari (masalan, Azure AI Foundry) 2026-yilda rivojlanayotgan ekotizimni qo'llab turadi ¹²⁵. Foundry agent service — bu AI agentlarni yaratish, deploy qilish va monitoring qilish uchun mo'ljallangan boshqaruv platformasi ²²². Bu, ayniqsa, korporativ sifatida ishlash uchun mo'ljallangan bo'lib, agent orchestrations, observability va secure model operations kabi xizmatlarni taklif etadi ¹¹⁰. Gemini API bilan integratsiya qilish uchun Azure AI Foundry yoki oddiy REST API orqali so'rovlar jo'natish mumkin. Azure AI Foundry'ning narxlarini baholash murakkab, chunki u agent platformasining narxi (oyiga 21–60/foydalanuvchi) va tokenlar uchun alohida to'lovni o'z ichiga oladi ¹¹³. Lekin bu platforma korporativ sifatida ishlash uchun mo'ljallangan bo'lib, agentni sozlash, deploy qilish va monitoring qilish uchun maxsus vositalarni taklif qiladi ²²².

Supabase va Render'ning hamjumladagi ekotizimi loyihalarni tez start qilish uchun ajoyibdir. Supabase'ning Auth va Storage kabi xizmatlari Render bilan integratsiyasi juda oson va minimal konfiguratsiya bilan ishlaydi ⁴⁸. Biroq, bu erkinlik narxini to'lashni talab qiladi: vendor lock-in, cheklangan resurslar va yuqori narxlardan keyin (masalan, Pro plan \$25/ora) ko'tarilishi ¹⁶⁸. Shuning uchun, loyiha kattalashganda, ba'zi komponentlarni boshqa platformaga ko'chirish (masalan, AI agent logic'ni Azure Functions'ga) haqidagi arxitekturaviy qarorlar qabul qilish kerak.

Masalan, 10,000 ta DAU (kunlik faol foydalanuvchi) darajasida yuklama oshishi bilan birga, Supabase'ning bog'lanishlar oqimida saqlash cheklavlari (60 ta bog'lanish) tugashi mumkin ²⁷. Bu muammoni hal qilish uchun Supabase'ni yoqish yoki Supabase'ning Pro planiga o'tish kerak bo'ladi ⁸⁹. Telegram bot'lar uchun Render'ning sleep problemi hal qilinmasa, bot'ni Cloudflare Workers'ga ko'chirish yoki Render'da "Always On" rejasini

sozlash kerak bo'ladi ¹²². AI agent'lar uchun yuklama oshganda, Gemini API'ning rate limits'iga tegish mumkin ¹¹. Bu muammoni hal qilish uchun so'rovlar oqimini boshqarish (queuing) va batch processing texnikalaridan foydalanish kerak. Masalan, Render Workflows yoki Azure Logic Apps kabi vositalardan foydalanib, so'rovlar oqimini boshqarish mumkin ²¹⁴²³⁶.

Xarajatlar bo'yicha, 100 dan 100,000 gacha foydalanuvchilarga o'tish jarayoni quyidagicha bo'lishi mumkin:

- **100–1,000 foydalanuvchi:** Supabase Free Plan + Render Hobby plan kombinatsiyasi deyarli xarajatsiz ishlaydi ^{26 114}.
- **1,000–10,000 foydalanuvchi:** Agar cheklovlar tugasa, Supabase Pro planiga (¹²⁰) va Render Pro planiga (¹¹⁷) o'tish kerak bo'ladi. Har biri *25/ora, demak, umumiy xarajat 50/ora* bo'ladi.
- **10,000+ foydalanuvchi:** Bu darajada cheklovlar chekka yetishi ehtimoli yuqori. Shu sababli, AI agent'lar uchun Azure Functions'ga ko'chirish, ma'lumotlar bazasi uchun Supabase'ni PostgreSQL on Azure'ga ko'chirish yoki storage uchun Cloudflare R2'ni premium turi (Infrequent Access) bilan ishlatish kabi optimallashtirish strategiyalarini qo'llash kerak bo'ladi ¹⁵⁷.

Xulosa qilib aytganda, Render'ning hamjumladagi ekotizimi loyihalarni tez start qilish uchun ajoyibdir, lekin u 10,000 dan ortiq foydalanuvchiga o'tish paytida cheklovlar bilan duch kelishi mumkin. Azure esa biroz murakkabroq, lekin potentsial ravishda bardoshliroq va ancha katta miqdorda foydalanuvchilarga yetkazish imkonini beradi. Loyihaning skalirlanish strategiyasini tuzishda bu ikki platforma orasida arxitekturaviy qarorlar qabul qilish muhimdir. Vendor lock-in riskini minimallashtirish uchun API'lar va xizmatlar orasida standartlarni qo'llash va ma'lumotlarni migratsiya qilish jarayonini oldindan tayyorlash tavsiya etiladi.

Cloudflare: Frontend, CDN va Ob'ekli Saqlash Arxitekturasi

Cloudflare platformasi 2026-yilda front-end ilovalar, CDN va ob'ekli saqlash uchun eng kuchli va xarajatli alternativlardan biri sifatida o'zini namoyish etmoqda. Ushbu bo'limda Cloudflare Pages, Cloudflare R2 va Cloudflare Workers kabi xizmatlar chuqur tahlil qilinadi. Ular Telegram bot'larini joylashtirish, statik va dinamik ilovalarni host qilish,

fayllarni saqlash va sozlash bo'yicha arxitekturaviy qarorlar, xarajatlar va cheklovlar bayon etiladi.

Cloudflare Pages — bu React, Vue, Svelte kabi modern JavaScript framework'larini avtomatik build va deploy qilish uchun mo'ljallangan platforma. Uning asosiy afzalliklari tezlik, global CDN's'ga kirish va avtomatik SSL sertifikatlaridir. Pages'ning bepul rejimi 500 ta preview deployment'ni oyiga o'z ichiga oladi, bu esa jamoaviy ishlash uchun juda foydalidir ⁴⁴. Preview deployment'lar, har bir pull request uchun avtomatik ravishda yaratiladigan va o'zingizning pull requestingizni sinab ko'rishingiz mumkin bo'lgan ilovalar. Bu, CI/CD jarayonini osonlashtiradi. Pages'ning cheklovlari esa kamroq, lekin 500 ta deploys'ga cheklov borligi va 500 MB'lik build cache hajmi e'tiborga loyiqdir. Agar loyihangiz juda katta bo'lsa (masalan, 500 dan ortiq preview deployment'lar kerak bo'lsa), bu rejim cheklovga urishi mumkin.

Cloudflare R2 — bu AWS S3'ga muvofiqlashtirilgan, kuchli, ishonchli va, ayniqsa, narx jihatidan ajoyib ob'ektlilik saqlash xizmatidir. R2'ning eng katta afzalligi — uni egress bandwidth'ining haddan tashqari yuqori bo'lishi bilan bog'liq. Bu, AWS S3 kabi xizmatlardan farqli ravishda, ma'lumotlarni yuklab olish uchun hech qanday to'lov qilmaslikni ta'minlaydi ¹⁶⁰¹⁶¹. R2'ning bepul rejimi 10 GB hajmdagi saqlash, 1 million Class A (metadata) va 10 million Class B (data) operatsiyalarini o'z ichiga oladi ¹⁶³. Bu rejim, birinchi navbatda, sinov uchun yoki juda kichik loyihalar uchun yetarli. R2'ning narxlari ham juda raqobatchilikka cho'zmaydigan. Standard saqlash uchun 1 GB oyiga taxminan 0.015, *infrequent access* uchun esa 0.01 ga teng ¹⁵⁷¹⁵⁸. R2'ning xarajatlarini kamaytirish uchun, turli xil ma'lumotlarni turli xil saqlash stratifikasida saqlash muhim. Masalan, foydalanuvchi profil rasmlari va kichik media fayllari uchun R2'ning standard saqlashini, ommaviy ma'lumotlar bazalari va arxivlar uchun esa *infrequent access*'ni ishlatish. Cloudflare R2'ning xarajatlarini kamaytirish bo'yicha optimallashtirish yo'llari quyidagilarni o'z ichiga oladi:

- **Turli xil ma'lumotlar uchun turli xil stratifikatsiya:** Profile rasmlari, media fayllari, arxivlar va boshqa ma'lumotlarni turli xil saqlash stratifikasida saqlash.
- **Presigned URLs:** Fayllarni yuklash va yuklab olish uchun presigned URL'lar yaratish. Bu, foydalanuvchilarga direktsiz kirish imkonini beradi, bu esa backend serverlarni yukini kamaytiradi.
- **R2 Local Uploads:** R2'ning bu xizmati, upload jarayonini 75% ga qisqartirishga yordam beradi, chunki u ma'lumotlarni yaqinroq joyga yozadi va uni asinxron ravishda asl joyga nusxalaydi ²³⁴.

Cloudflare Workers — bu serverless funksiyalar platformasi bo'lib, u yerda JavaScript, TypeScript, Rust kabi tillarda yozilgan kodni global Cloudflare tarmog'ida ishlatish

mumkin. Telegram bot'larini joylashtirish uchun Cloudflare Workers juda maqbul variantdir, chunki u "sleep" funksiyasiga ega emas va har doim ham faol bo'ladi. Bu esa bot'ning tez vaqtida javob berishini ta'minlaydi. Workers'ning bepul rejimi 100,000 ta har kunlik so'rovni va 10,000 Neuron har kunlik so'rovni o'z ichiga oladi ³⁷ ⁴³. Bu rejim, bir nechta kichik bot'lar yoki sinov uchun yetarli. Workers'ning cheklovlari esa so'rovlar soni va CPU vaqtidir. Agar so'rovlar soni cheklovga yetib boring, xizmatni to'liq to'langan reja'ga ko'chirish kerak bo'ladi ¹⁵⁶.

Cloudflare D1 — bu serverless SQLite ma'lumotlar bazasi xizmatidir. Uning bepul rejimi 5 GB hajmdagi ma'lumotlarni saqlashni va D1'ning qisman narxli bo'lgan 2026-yil sentyabr holatida 5 GB'lik limitini o'z ichiga oladi ¹⁹⁷¹⁹⁸. D1'ning narxi satirlantiruvchi, chunki u qat'iy limitlar bilan ishlaydi. September 1, 2026 dan boshlab, D1'dagi row read/write limitlariga tegish bo'ldi va query'lar shunga bog'liq bo'lib qoldi ⁹⁸. Durable Objects esa per-person stateful ob'ektlarni yaratish uchun mo'ljallangan. Ularni Telegram bot'larini joylashtirish uchun foydalanish mumkin, masalan, har bir foydalanuvchi uchun alohida Durable Object yaratib, uning holatini saqlash ⁶⁶.

Supabase Storage va Cloudflare R2'ni taqqoslashda, ikkalasi ham ma'lumotlarni saqlash uchun mo'ljallangan, lekin har biri o'zining afzalliklariga ega. Supabase Storage'ning afzalliklari uning S3 kompatibil bo'lishi va uning supabase.org interfeysida boshqarilishi. Bu, ayniqsa, Supabase'ning boshqa xizmatlariga (Auth, Database) yaqqol integratsiyani talab qiladigan loyihalar uchun maqbul. Lekin Supabase'ning bepul rejimining 1 GB'lik cheklovi va 5 GB'lik egress bandwidth'iga cheklovi mavjud ¹⁶⁵. Cloudflare R2'ning afzalliklari esa uning "zero egress fee" modeli, 10 GB'lik bepul rejim hajmi va uni 5 GB'lik bepul rejim hajmi bilan bog'liq cheklovlari mavjud ¹⁶³. Shuning uchun, agar loyiha foydalanuvchi tomonidan yuklab olinadigan katta hajmdagi fayllar (rasmlar, videolar) ko'paytirsang, R2'ning narx modeli ancha samaraliroq bo'ladi.

Quyidagi jadvalda Cloudflare xizmatlari taqqoslanadi:

Xizmat	Asosiy Ijro	Bepul Rejim Limitlari (2026-yil sentyabr)	Xarajatlar (Pay-Per-Use)	Optimal Foydalanish
Cloudflare Pages	Statik veb-illovalar	500 ta preview deployment'lar	Pay-Per-Use	React, Vue, Svelte ilovalari
Cloudflare R2	Ob'ektlilik saqlash	10 GB saqlash, 1M Class A, 10M Class B operatsiyalar	\$0.015/GB/ora (Standard) ¹⁵⁷	Media fayllar, user uploads, backups
Cloudflare Workers	Serverless funksiyalar	100,000 ta so'rov/kun, 10,000 Neuron/kun ³⁷	\$9.00/million so'rov ³⁶	Telegram bot'lar, API gateway'lar, webhook handlers
Cloudflare D1	Serverless SQLite	5 GB saqlash ¹⁹⁸	\$0.75/GB-oradan keyin ¹⁹⁷	Kichik ma'lumotlar bazalari, personal data
Cloudflare Durable Objects	Stateful ob'ektlar	500 ta ob'ektlar (Free plan'da)	Additive to Workers pricing ⁸⁵	Per-user state, game servers, queues

Xulosa qilib aytganda, Cloudflare platformasi 2026-yilda front-end, storage va compute uchun kuchli va xarajatli alternativa taklif etadi. Cloudflare Pages front-end ilovalar uchun ideal, R2 esa media fayllar uchun narx jihatidan ajoyib, Workers esa Telegram bot'larini joylashtirish uchun eng maqbul variantdir. Ushbu xizmatlarni aralash qo'llash orqali, loyihaning xarajatlarini minimal darajada saqlab, ishonchligini va tezligini oshirish mumkin.

Telegram Bot va Mini Apps: Integratsiya, Xavfsizlik va Erkinliklar

Telegram Bot va Mini Apps'lar 2026-yilda AI agent'lar va boshqa murakkab ilovalarni Telegram platformasi ichida ishga tushirish uchun asosiy vositalar hisoblanadi. Ushbu bo'limda ularning integratsiyasi, xavfsizlik mexanizmlari va erkinliklar tahlil qilinadi. Ushbu tahlil senior dasturchilar uchun Telegram bot'larini va Mini Apps'larini ishonchli va xavfsiz tarzda qanday qurish, boshqarish va optimallashtirish mumkinligini ko'rsatadi.

Telegram Bot API'ning asosiy ikki usuli mavjud: webhook va long polling. Prodakshen muhitida webhook'lar yuqori yuklamalar uchun ancha maqbuldir. Webhook'lar orqali Telegram serveringizning endpoint'iga ma'lumotlar jo'natiladi, bu esa serverlarni har doim so'rovlar uchun tekshirib turishni yo'qotib, resurslarni tejashga yordam beradi. Long polling esa, bot'ning HTTP serveri ishlamasa, Telegram serveri so'rovni o'zida saqlab turadi va javob beradi. Bu, ayniqsa, bot'ni ishga tushirish va sinovdan o'tkazish uchun foydalidir. Lekin, Telegram bot'lar uchun long polling'ni ishlatish tavsiya etilmaydi, chunki bot serveri ishlamasligi mumkin va so'rovlar yo'qotilishi mumkin. Shuning uchun,

Telegram bot'larini joylashtirish uchun avtomatik deploy va ishonchli server (masalan, Render, Cloudflare Workers) talab etiladi. Bot'ni joylashtirishda eng katta muammolardan biri Render'ning "sleep" funksiyasidir. Agar bot 15 daqiqa davomida hech qanday so'rov qabul qilmasa, Render konteynerini to'xtatadi va birinchi so'rov kelgach, konteyner qayta ishga tushadi ¹²². Bu "cold start" jarayoni bir necha soniyadan bir necha daqiqagacha davom etishi mumkin, bu esa bot'ning real vaqt rejasida javob berish talablariga zid. Shuning uchun, Telegram bot'larini joylashtirish uchun Render'da "Always On" rejasini sozlash yoki bot'ni Cloudflare Workers kabi boshqa xizmatga ko'chirish tavsiye etiladi ⁶⁷.

Telegram Mini Apps'lar uchun xavfsizlik — bu eng muhim masala. Mini App'lar Telegram ichida ishlaydigan web ilovalar bo'lib, ular `initData` nomli parametr orqali Telegram tomonidan ma'lumotlarni o'zlariga jo'natadi. `initData`'dagi hash qiymati, bot'ning `secret_key` yordamida hisoblanadi. `initData`'ni frontend'da ishonchli deb hisoblash xavfli. Hashni serverda avtomatlashtirilgan validatsiya algoritmi orqali tekshirish kerak. Validatsiya jarayoni quyidagilarni o'z ichiga oladi: 1. `initData`'dagi barcha kalitlarni (`chat_type`, `user`, `start_param` va h.k.) `initDataUnsafe`'dan oling va alfavit bo'yicha tartiblang. 2. Kalitlar va qiymatlari birlashtiriladi va stringga aylantiriladi. 3. Bu stringga bot'ning `secret_key` qo'shiladi. 4. Hosil bo'lgan matnga HMAC-SHA256 algoritmi qo'llaniladi va natija hisoblanadi. 5. Hosil bo'lgan hash qiymati `initData`'dagi hash qiymati bilan solishtiriladi. Agar ikkita hash mos kelmasa, so'rov noma'lum manbadan kelgan va rad etilishi kerak.

Node.js yoki Python'da bu validatsiya algoritmini quyidagicha amalga oshirish mumkin. Bu validatsiya serverda har doim amalga oshirilishi kerak va bot'ning `secret_key` hech qachon public joylarda ko'rinmasligi lozim.

Telegram bot'lariga tegishli erkinliklar ham muhimdir. Telegram API'ning har bir API method'iga tegishli erkinliklar mavjud. Misol uchun, `/start` kabi command'lar uchun global erkinliklar mavjud, lekin biron bir chat'da so'rovlar uchun ham chat-specifik erkinliklar mavjud. So'rovlar chegarasini oshirish uchun Google AI Studio'da so'rov qo'shish mumkin ²²⁴. Agar so'rovlar chegarasini oshirish mumkin bo'lmasa, so'rovlar oqimini boshqarish (queuing) va exponential backoff bilan qayta urinish mexanizmlaridan foydalanish kerak ²¹⁷.

Quyidagi jadvalda Telegram bot'larida uchraydigan eng keng tarqalgan hatoliklar va ularni hal qilish yo'llari keltirilgan:

Hatolik Kod	Nomlanishi	Mumkin bo'lgan Sabab	Diagnostika	Hal qilish	Prevenziya
400	Bad Request	Noto'g'ri so'rov formati, yaroqsiz parametrlar	So'rov JSON'ini tekshirish, Telegram API dokumentatsiyasiga moslashtirish	So'rovni to'g'rilash va qayta jo'natish	Avvaldan so'rovni validatsiya qilish
401	Unauthorized	Noto'g'ri yoki eskirgan bot token'i	Bot tokenini Tekshirish	Bot tokenini yangilash va qayta sozlash	Tokenni .env fayli va environment variables'da saqlash
403	Forbidden	Bot'ning API'ga kirish huquqi yo'q, bot blocklangan	BotFather orqali bot holatini tekshirish	Blocklangan bo'lsa, unblock qilish yoki yangi bot yaratish	Tokenni serverda faqat kerakli joylarda foydalanish
429	Too Many Requests	Rate limitlariga tegish	Retry-After header'ini tekshirish	So'rovlar sonini kamaytirish yoki exponential backoff bilan qayta urinish	So'rovlar oqimini boshqarish (queuing), erkinliklar oqimini nazorat qilish
Timeout	Connection timed out	Internet tarmog'ida muammolar, server ishlamayapti	Server holatini tekshirish, ping/pong testlari	Server holatini yaxshilash, retry mechanism'ni qo'llash	Isbor server, DNS sozlamalari, SSL sertifikatlarini tekshirish

Telegram Mini App'larini integratsiya qilishda, foydalanuvchi interfeysi sifatida Cloudflare Pages'da joylashtirilgan React/Vue ilovasi ishlatiladi. Mini App'ning `index.html` fayli Telegram'ning Telegram.WebApp API'sini yuklaydi. Telegram bot'ga `start_param` bilan havola jo'natish orqali Mini App'ni ochish mumkin. Bu, bot'lar uchun onboarding, deep linking va notification'lar uchun kuchli vosita bo'lib xizmat qiladi. Masalan, bot'ning "Dashboard" tugmasini bosganda, Mini App'ni ochadigan havola jo'nating. Mini App'ning ichida esa, foydalanuvchi ma'lumotlarini, chat ma'lumotlarini va boshqa kerakli ma'lumotlarni `Telegram.WebApp.initData` orqali olish mumkin. Shundan so'ng, bu ma'lumotlar backend serveriga jo'natiladi va serverda `initData`'ni validatsiya qilinadi.

Xulosa qilib aytganda, Telegram bot'larini va Mini Apps'larini integratsiya qilishda, webhook'lar prodakshen muhitida ishlatilishi, bot'ni ishonchli joylashtirish (masalan, Cloudflare Workers), `initData`'ni serverda validatsiya qilish va hatoliklarni to'g'ri boshqarish muhimdir. Ushbu amaliy yo'llar yordamida, Telegram platformasi ichida ishonchli, xavfsiz va ishonchli AI agent'lar va ilovalar yaratish mumkin.

Gemini API: AI Agent Arxitekturası, Token Optimallashtirish va Narxlantirish

Gemini API 2026-yilda AI agent'lar uchun asosiy hisoblanadi va murakkab tahlil, kod yozish va multimodal operatsiyalar uchun kuchli vosita bo'lib qolmoqda ¹⁴⁴. Ushbu bo'limda Gemini API'ning turli modellari, narxlari va erkinliklar tahlil qilinadi. Shuningdek, tokenlarni tejash strategiyalari (prompt caching, summarization, embeddings) va AI agent arxitekturası (memory, tool calling, planning) kabi muhim masalalar chuqur ko'rib chiqiladi.

Gemini API ekotizimida turli xil model turlari mavjud bo'lib, har biri o'zining narxi va imkoniyatlari bilan ajralib turadi. `gemini-3.8-flash` modeli, 1,048,576 token'li context window'ga ega bo'lgan tezkor va arzon flash model sifatida e'lon qilingan ¹⁵⁰. Bu model, tezkor javoblarni talab qiladigan interaktiv agent'lar uchun ideal. `gemini-3.5-pro` esa murakkab tahlil, kod yozish va sloppiy tushunchalarni tushunish uchun mo'ljallangan ²²⁸. `gemini-omni-1.1-flash` esa video generatsiyasi va editing uchun mo'ljallangan ²²⁹. Loyiha uchun "model routing" strategiyasi yaratish kerak: oddiy so'rovlar uchun Flash, murakkab tahlil uchun Pro va multimodal operatsiyalar uchun Omni kabi modelni tanlash. Model tanlashda narx ham muhim omil. Masalan, `gemini-2.5-flash-lite` modelining narxi taxminan 1 million input token uchun 0.10 va output token uchun 0.40 hisoblanadi ⁷⁴. Bundan tashqari, Gemini API'ning Live API modeli bidirectional audio va video uchun mo'ljallangan, lekin uning erkinliklari yanada cheklangan ⁴.

Gemini API'ning xarajatlari asosan tokenlar soniga bog'liq. Tokenlarni tejash strategiyalari juda muhim, ayniqsa, chat history kattalashganda. Quyidagi optimallashtirish usullaridan foydalanish mumkin:

- **Prompt Caching:** Gemini API'ning prompt caching imkoniyatidan foydalanish. Agar so'rov avvalroq takrorlanib ketsa, u qayta hisoblanmaydi va tezroq javob beradi.
- **Context Caching:** So'rovlar oqimini boshqarish uchun queuing texnikalaridan foydalanish. Agar so'rovlar chegarasini oshirish mumkin bo'lmasa, so'rovlar oqimini boshqarish (queuing) va exponential backoff bilan qayta urinish mexanizmlaridan foydalanish kerak ²¹⁷.
- **Summarization:** Chat history'ni davomiy ravishda yig'ib borish olib kelishi mumkin bo'lgan token sarfini kamaytirish uchun summarization'ni qo'llash. Masalan, har 5 ta so'rovdan keyin chat history'ni qisqacha qilib yozish.
- **Embeddings and RAG (Retrieval-Augmented Generation):** Foydalanuvchi so'rovlari va chat history'ni embeddings'ga aylantirib, vector database'da saqlash.

So'rov kelganda, eng yaqin embeddings'larni topib, ularni context sifatida Gemini API'ga jo'natish. Bu, ayniqsa, katta hajmdagi ma'lumotlar bazasidan ma'lumotlarni izlash uchun samarali.

- **Model Routing:** Har bir so'rovning murakkabligiga qarab turli modellar ishlatish. Oddiy so'rovlar uchun arzon model, murakkab tahlil uchun qimmat model.

AI agent arxitekturasi murakkabroq bo'lib tuyuladi. U quyidagi komponentlardan iborat:

- **System Prompt:** Agent'ning vazifasini, xatti-harakatlarini va o'zini-ta'kidlashini belgilaydigan initial instructions.
- **Conversation History:** Agent'ning foydalanuvchi bilan o'tkazgan suhbatlarini saqlaydigan buffer.
- **Memory:** Agent'ning qisqa muddatli (conversation history) va uzun muddatli (vector database'da saqlangan embeddings) xotirasini boshqarish.
- **Tool Calling and Function Calling:** Agent'ga tashqi xizmatlarni (masalan, Google Search, API'lar) chaqirish imkonini beruvchi Gemini API'ning imkoniyati.
- **Planning and Agent Loops:** Agent'ning murakkab vazifani oddiy qadamlarga bo'lishi va ularni ketma-ket bajarydigan planning mexanizmini boshqarish.
- **Background Jobs:** Agent'ning so'rovlarini qayta urinishi, validatsiyasi va boshqa xizmatlarni chaqirishi uchun background jobs (masalan, Render Workflows) yoki queues (masalan, RabbitMQ, Redis) ishlatish.

Quyidagi jadvalda Gemini API'ning 2026-yil holatidagi asosiy model turlari va narxlari keltirilgan:

Model Nomi	Context Window	Output Limit	Input Narxi (per 1M tokens)	Output Narxi (per 1M tokens)
Gemini 3.8 Flash	1,048,576 token	-	0.75 3.75 ⁶	
Gemini 3.5 Flash Lite	65,536 token	65,536 token	0.30 2.50 ¹⁰	
Gemini 2.5 Flash-Lite	65,536 token	65,536 token	0.10 0.40 ⁷⁴	
Gemini 3.1 Pro	2,000,000 token	8,000,000 token	2.00 18.00 ²³²	
Gemini Omni Flash	-	-	-	- (Video Generatsiya) ²²⁹

Gemini API'ning erkinliklari ham muhimdir. Erkinliklar mavjud bo'lib, ular turli usullar va tizimlar uchun farq qiladi ^{72 253}. So'rovlar chegarasini oshirish uchun Google AI Studio'da so'rov qo'shish mumkin ²²⁴. Agar so'rovlar chegarasini oshirish mumkin bo'lmasa, so'rovlar oqimini boshqarish (queuing) va exponential backoff bilan qayta urinish mexanizmlaridan foydalanish kerak ²¹⁷. So'rovlar chegarasini oshirish uchun Google AI Studio'da so'rov qo'shish mumkin ²²⁴.

Xulosa qilib aytganda, Gemini API 2026-yilda AI agent'lar uchun juda kuchli vosita bo'lib qolmoqda. Turli xil model turlari, narxlari va erkinliklar mavjud. Loyiha uchun "model routing" strategiyasi yaratish, tokenlarni tejash uchun optimallashtirish usullaridan foydalanish va AI agent arxitekturasini memory, tool calling va planning kabi komponentlar bilan jihozlash muhimdir. Ushbu yondashuvlar yordamida, samarali, xarajatli va ishonchli AI agent'lar yaratish mumkin.

Yakuniy Arxitektura, Xarajat Boshqaruvi va Mas'uliyatli Dasturlash Praktikasi

Ushbu tadqiqotning yakuniy qismida, taqdim etilgan texnologiyalar va tahlillar asosida loyihaning xarajatlarini nazorat qilish, mas'uliyatli dasturlash praktikalarini qo'llash va yakuniy arxitekturaviy qarorlarni qabul qilish bo'yicha yakuniy maslahatlar taqdim etiladi. Ushbu qismda loyihaning xarajatlarini minimal darajada saqlash uchun turli xil variantlar (deyarli xarajatsiz, 5,25, \$50 va masshtablantirilgan) keltiriladi. Shuningdek, xavfsizlik, debugging va backup kabi muhim sohalar bo'yicha amaliy maslahatlar ham beriladi.

Loyihaning xarajatlarini nazorat qilish uchun quyidagi arxitekturaviy variantlar taklif etiladi. Har bir variantda asosiy komponentlar (Frontend, Backend, Database, Storage, AI, etc.) tanlangan texnologiyalar orqali amalga oshiriladi.

1. Zero-Cost Architecture (\$0/month): Bu arxitektura deyarli xarajatsiz, sinov uchun yoki juda kichik loyihalar uchun mo'ljallangan. Uning cheklovlari mavjud, lekin u loyihani boshlash uchun eng arzon usuldir.

- **Frontend:** Cloudflare Pages (bepul reja) ⁴⁴.
- **Backend:** Render Hobby (bepul reja) ¹¹⁴.
- **Database:** Supabase Free Plan (500 MB DB, 1 GB Storage) ²⁶.
- **Auth:** Supabase Auth (bepul reja) ¹⁶⁵.
- **Storage:** Supabase Storage (1 GB) ¹⁶⁵.
- **Bot Hosting:** Render Hobby (bepul reja) ¹¹⁴.
- **AI:** Gemini API free tier (rate limits mavjud) ⁶.
- **Monitoring:** Grafana Cloud free tier (Supabase bilan integratsiya) ²⁶⁸.
- **Xarajatlar:** Umumiy xarajat deyarli \$0, lekin cheklovlar (egress, storage, connections) tez tinchaydi.

2. \$5/month Architecture: Bu arxitektura cheklovlar bilan bog'liq muammolarni bartaraf etish uchun yuqori darajadagi resurslarni taqdim etadi.

- **Frontend:** Cloudflare Pages (bepul reja) 44.
- **Backend:** Render Starter (taxminan \$7/ora, lekin free reja bilan ishlaydi) 117.
- **Database:** Supabase Pro Plan (\$25/ora) 120.
- **Storage:** Cloudflare R2 (standard) 157.
- **Bot Hosting:** Cloudflare Workers (bepul reja) 37.
- **AI:** Gemini API (free tier) 6.
- **Monitoring:** Grafana Cloud free tier 268.
- **Xarajatlar:** Taxminan $25/ora(SupabasePro) + 5/ora (Cloudflare R2) = \sim \$30/ora$.
Bu narxni pasaytirish uchun R2'ning infrequent access'ini qo'llash mumkin 157.

3. \$25/month Architecture: Bu arxitektura 1,000-10,000 foydalanuvchilar uchun mo'ljallangan va cheklovlar bilan bog'liq muammolarni bartaraf etish uchun yuqori darajadagi resurslarni taqdim etadi.

- **Frontend:** Cloudflare Pages (bepul reja) 44.
- **Backend:** Render Pro (\$25/ora) 117.
- **Database:** Supabase Pro Plan (\$25/ora) 120.
- **Storage:** Cloudflare R2 (infrequent access) 157.
- **Bot Hosting:** Cloudflare Workers (bepul reja) 37.
- **AI:** Gemini API (free tier) 6.
- **Monitoring:** Grafana Cloud free tier 268.
- **Xarajatlar:** Taxminan $25/ora(RenderPro) + 5/ora (Cloudflare R2) = \sim \$30/ora$.
Narxni pasaytirish uchun R2'ning infrequent access'ini qo'llash mumkin 157.

4. \$50/month Architecture: Bu arxitektura 10,000 dan ortiq foydalanuvchilar uchun mo'ljallangan va yuqori yuklamalarni qabul qilish uchun mo'ljallangan.

- **Frontend:** Cloudflare Pages (bepul reja) 44.
- **Backend:** Render Pro (\$25/ora) 117.
- **Database:** Supabase Pro Plan (\$25/ora) 120.
- **Storage:** Cloudflare R2 (standart) 157.
- **Bot Hosting:** Cloudflare Workers (bepul reja) 37.
- **AI:** Gemini API (free tier) 6.
- **Monitoring:** Grafana Cloud free tier 268.
- **Xarajatlar:** Taxminan \$50/ora (Render Pro + Supabase Pro).

Mas'uliyatli dasturlash praktikalari loyihaning ishonchliligini va xavfsizligini ta'minlaydi. Quyidagi checklist'ni bajarish tavsiya etiladi:

- **Secrets Management:** Barcha API kalitlari, tokenlar va boshqa sirli ma'lumotlar `.env` fayllarida yoki kodda saqlanmasligi kerak. Ular GitHub Actions'da `GitHub Secrets` ²¹³ va Render/Azure'da mantiqiy ravishda ulanadigan environment variables yordamida boshqarilishi lozim. Kalitlar davriy ravishda almashtirilishi (rotate) va agar ular Git tarixida ochiq joylashtirilsa, tezda bekor qilinishi kerak ⁹⁶.
- **Input Validation:** Foydalanuvchi kiritgan barcha ma'lumotlarni serverda to'g'ri va xavfsizligini tekshirish. Bu SQL injection, XSS kabi hujumlarni oldini olishga yordam beradi.
- **Rate Limiting:** Bot'lar va API'lar uchun so'rovlar sonini cheklovlar bilan cheklash. Bu DoS hujumlarini va API'ning yolg'on foydalanishini oldini oladi.
- **Logging:** Loyihaning barcha muvaffaqiyatli va muvaffaqiyatsiz harakatlarini keng ko'lamli loglarni yig'ish. Loglarni markazlashtirib, masalan, Grafana'ga yuborish yoki Log Drains orqali tashqi manzilga jo'natish samarali yondashuv bo'lishi mumkin ¹⁸⁸.
- **Backup and Disaster Recovery:** Ma'lumotlar bazasi va R2'dagi ma'lumotlarni davriy ravishda backup qilish. RPO (Recovery Point Objective) va RTO (Recovery Time Objective) tushunchalarini tushunish va ularga mos recovery strategiyalari tuzish muhimdir.

Debugging va troubleshooting jarayonini osonlashtirish uchun debugging yo'riqnoma va error handbook'lar yaratish muhimdir. Ushbu qo'llanmada keltirilgan 100 dan ortiq muammolarni hal qilish bo'yicha tafsilotli yo'riqnoma mavjud ^{50 51}.

Xulosa qilib aytganda, 2026-yil holatida Telegram bot, Mini App va AI agent'lar uchun ishonchli va xarajatli arxitektura yaratish mumkin. Ushbu arxitektura, Supabase, Render, Cloudflare va Gemini API kabi texnologiyalarni aralash qo'llash orqali minimal xarajat bilan maksimal foydalanuvchiga xizmat ko'rsatishga yordam beradi. Loyihani boshlashdan oldin va davomida xavfsizlik, debugging va cost optimization kabi masalalarni hisobga olish muhimdir. Ushbu qo'llanma senior dasturchilar uchun loyihani noldan prodakshenga chiqarish uchun amaliy va zamonaviy yechimlarni taklif etadi.

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